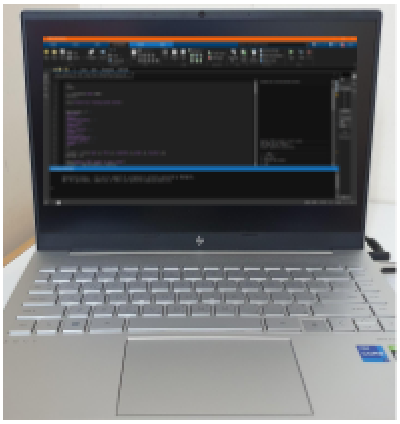
Name: Justin M. Iadicola Date: 5/11/2026 Class: 8

MyDesign Portfolio

Element D

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

Design Idea #1	
Date:	*4/10/2026*
Description:	*The first design idea was a basic MATLAB-only hall pass system using manual keyboard input. Teachers would type the student's name, destination, and allowed time limit. The program would use a timer to determine whether the student returned on time or late.*
Drawing that includes distinguishing information such as scale, dimensions, and materials:	 Computer Running First Iteration Script in MATLAB

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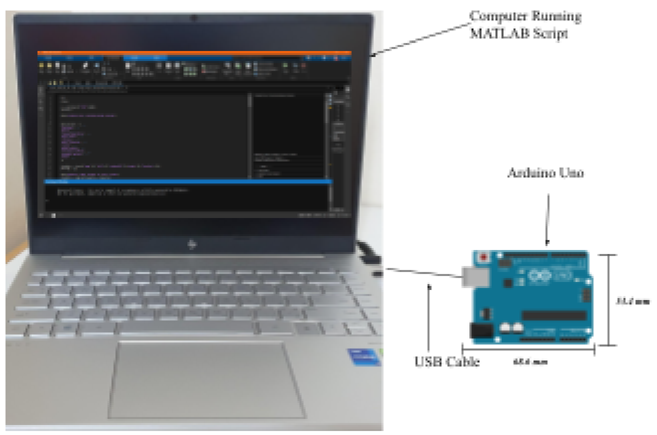
Design Idea #2	
Date:	*4/13/2026*
Description:	*The second design added Arduino integration with a push button sensor to simulate student arrival at a destination. MATLAB would receive a signal from the Arduino when the button was pressed, allowing the system to detect arrival automatically.*
Drawing that includes distinguishing information such as scale, dimensions, and materials:	

Design Idea #3	
Date:	*4/16/2026*
Description:	*The third design replaced the push button with an RFID scanner to uniquely identify students. Each student would scan an RFID tag when leaving and arriving at a destination. MATLAB would compare RFID data to stored student records.*

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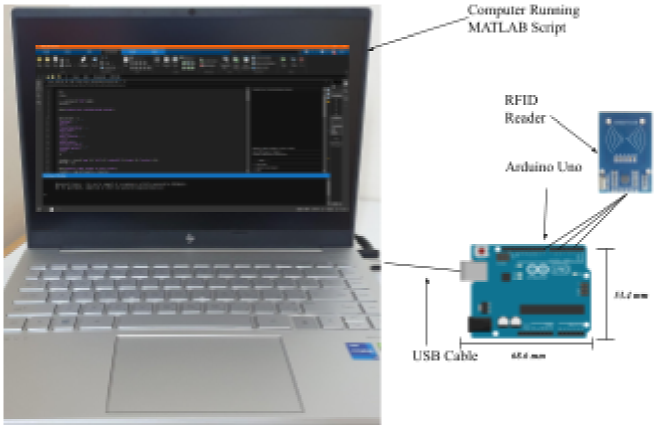
Drawing that includes distinguishing information such as scale, dimensions, and materials:	

Design Idea #4	
Date:	*4/19/2026*
Description:	*The fourth design introduced a student database system. Students could register their name, RFID tag, student ID, grade, and location into a MATLAB structure array. The system would allow multiple students to be stored and identified automatically.*

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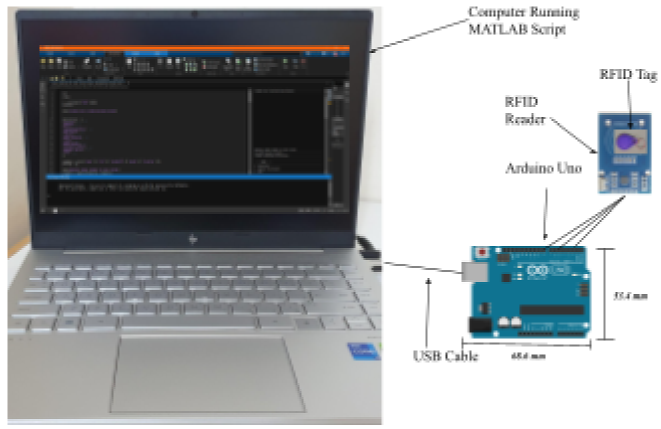
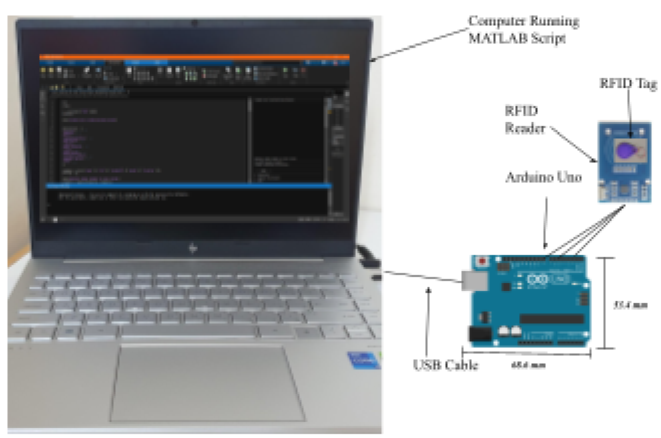
Drawing that includes distinguishing information such as scale, dimensions, and materials:	

Design Idea #5	
Date:	*4/22/2026*
Description:	*The final design added a destination menu system and input validation. Teachers select destinations from a predefined list while still supporting a custom “Other” option. Invalid inputs are detected and rejected automatically.*
Drawing that includes distinguishing information such as scale, dimensions, and materials:	

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Design Idea Comparison and Selection *(D1, D2)*

Requirement	Weight	Design 1	Design 2	Design 3	Design 4	Design 5
RFID Identification	5	1	2	5	5	5
Time Tracking	5	4	4	5	5	5
Student Database	4	1	1	2	4	4
Ease of Use	5	1	2	3	3	5
Input Validation	3	1	1	2	3	3
Real-Time Tracking	5	1	3	5	5	5
Total score		9	13	22	25	27

Our team chooses Design 5 to solve the student accountability and hall pass tracking problem because it fulfills all major design requirements, including RFID identification, time tracking, multiple student registration, input validation, and ease of use. Design 5 provides the most realistic and reliable solution while improving system stability and user experience.

Design Blueprint *(D4)*

Design Blueprint #1

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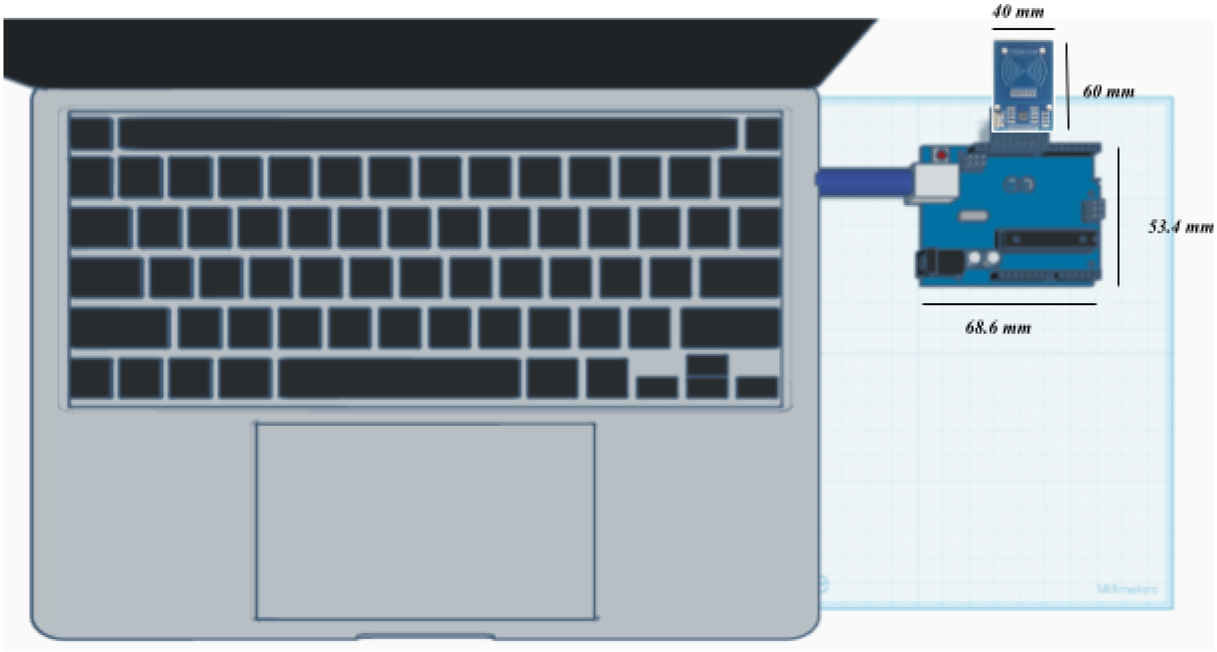
Date: 4/16/2026
Sketch or CAD Drawing (orthographic/multiview):



Manufacturing Method:

Step 1:

Connect MFRC522 RFID module to Arduino Uno using jumper wires and SPI pin configuration.

Step 2:

Upload Arduino RFID reading code using Arduino IDE.

Step 3:

Connect Arduino to MATLAB using serial communication and test RFID scanning functionality.

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Step 4:

Program MATLAB interface with student registration, destination tracking, and timing system.

Step 5:

Test system repeatedly with multiple RFID tags and destination scenarios.

Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**
Arduino Uno R3	School Provided	1	$0
MFRC522 RFID Module	School Provided	1	$0
RFID Cards/Tags	School Provided	1	$0
Arduino Base Station	School Provided	1	$0
Jumper Wires	School Provided	1	$0
USB Cable	School Provided	1	$0
Laptop with MATLAB	School Provided	1	$0
Total Cost			$0

Plan of Action *(D3)*

Week	**Task**

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Week 1	Research existing hall pass systems
Week 1	Create initial MATLAB-only prototype
Week 2	Integrate Arduino communication
Week 2	Implement RFID scanning functionality
Week 3	Develop student database system
Week 3	Add destination selection and validation
Week 4	Test prototype repeatedly
Week 4	Debug errors and improve reliability
Week 4	Record demonstration video
Week 4	Prepare final presentation

Iterations for the Solution:

Date of Iteration:

4/10/2026

Description of Iteration:

Created the initial MATLAB-only pass system using manual input. The system could record student names, destinations, and determine whether students returned on time or late using a timer.

Result:

The logic functioned correctly, but there was no method to verify student identity or actual arrival.

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Date of Iteration:

4/13/2026

Description of Iteration:

Integrated Arduino hardware using a push button sensor to simulate student arrival detection.

Result:

MATLAB successfully received signals from Arduino, improving realism, but the system still could not identify which student arrived.

Date of Iteration:

4/16/2026

Description of Iteration:

Added MFRC522 RFID scanner to uniquely identify students through RFID tags.

Result:

RFID communication worked successfully after filtering incorrect serial data and preventing empty scans from being accepted.

Date of Iteration:

4/19/2026

Description of Iteration:

Implemented a student registration database storing names, RFID tags, IDs, grades, and locations.

Result:

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The program became more organized and realistic while supporting multiple students.

Date of Iteration:

4/22/2026

Description of Iteration:

Added destination menus, custom “Other” destinations, and input validation for invalid entries.

Result:

Improved reliability, reduced user errors, and strengthened overall usability of the system.